CO-1 (AT-3)

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1. Smart home Automation System Analysis:

A smart home uses intelligent agents to sense its surroundings, Make decisions, and control devices such as lights, locks, alarms, and thermostats. Different agent types are:

- Simple Reflex Agent:

Acts only on the current percept using condition - action rules. eg: It switches on light when motion sensor detects movement. It is fast but does not remember past.

- Model-Based reflex Agent:

Maintains an internal state of the environment, including information that may not be directly observable. eg: It remembers that a door was locked before even checking.

- Goal-Based Agent:

Choose actions that help achieve a specific goal. eg: It adjust heating or cooling until the home reaches the user's preferred temperature of 22°C.

## Utility-Based Agent:

compares possible outcomes and selects the one with the greatest overall benefits. eg: It balance comforts, electricity use, and security when deciding how to operate home.

## Hybrid Intelligent Agent:

A hybrid agent combining reflex, model-based, goal based, and utility-based features is the most suitable choice. Reflex rules can respond immediately to dangers such as activating an alarm when a break-in is detected. Such as comfort and energy savings.

## Conclusion:

A hybrid approach is effective because it combines quick reactions with a memory and careful decision-making. This makes the smart home safer, more comfortable and more energy-efficient.

2) ## Unknown Maze Navigation & search strategies:

→ An Autonomous agent navigating an unknown maze must discover paths, avoid obstacles, remember visited junctions, and find a route to the exit without repeatedly exploring the same areas.

## Uninformed search strategies:

### • Uniform Cost Search (UCS):

Expands the node with the lowest accumulated path cost, g(n). when step cost are non-negative, It finds the least-cost path. It is useful when different routes have different costs, but it may use considerable time and Memory because it explores Many ethics.

### • Iterative Deepening Search (IDS):

Runs depth-limited search repeatedly, Increasing the depth limit each time. It combines the low memory use of depth-first search with completeness of breadth first search. with equal step costs, it finds a shallowest solution; however, nodes near the top may be Generated.

## Agent state & Rational Decisions:

A model-based agent is suitable because it maintains an internal map of the Maze, recording known walls, open paths, and visited junctions. After each move, it updates this info using new percepts. This helps it avoid loops and choose actions based on what it has already discovered.

## Conclusion:

The model-based agent provides memory of the Maze, while the search strategy provides a systematic way to select a route. Together, they support safer exploration and a cost-effective path to the exit.